Design Method

Digital clock MM:SS

- Four digit counters: seconds units (0-9), seconds tens (0-5), minutes units (0-9), minutes tens (0-5)
- One 1 Hz Clock signal drives all counters
- Each digit is shown on a hex 7-segment display
- A higher digit counts only when all lower digits are at their maximum

Digit counter block

- 4-bit register stores the digit
- Adder computes Q + 1 on every clock tick
- Comparator checks Q = Max
- If equal, the multiplexer loads 0 instead of Q + 1
- Output wrap = (Q = Max) is the carry to the next digit

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Truth Tables

Units counter (Max = 9)

Q	Next Q	wrap
0	1	0
1	2	0
2	3	0
3	4	0
4	5	0
5	6	0
6	7	0
7	8	0
8	9	0
9	0	1

Tens counter (Max = 5)

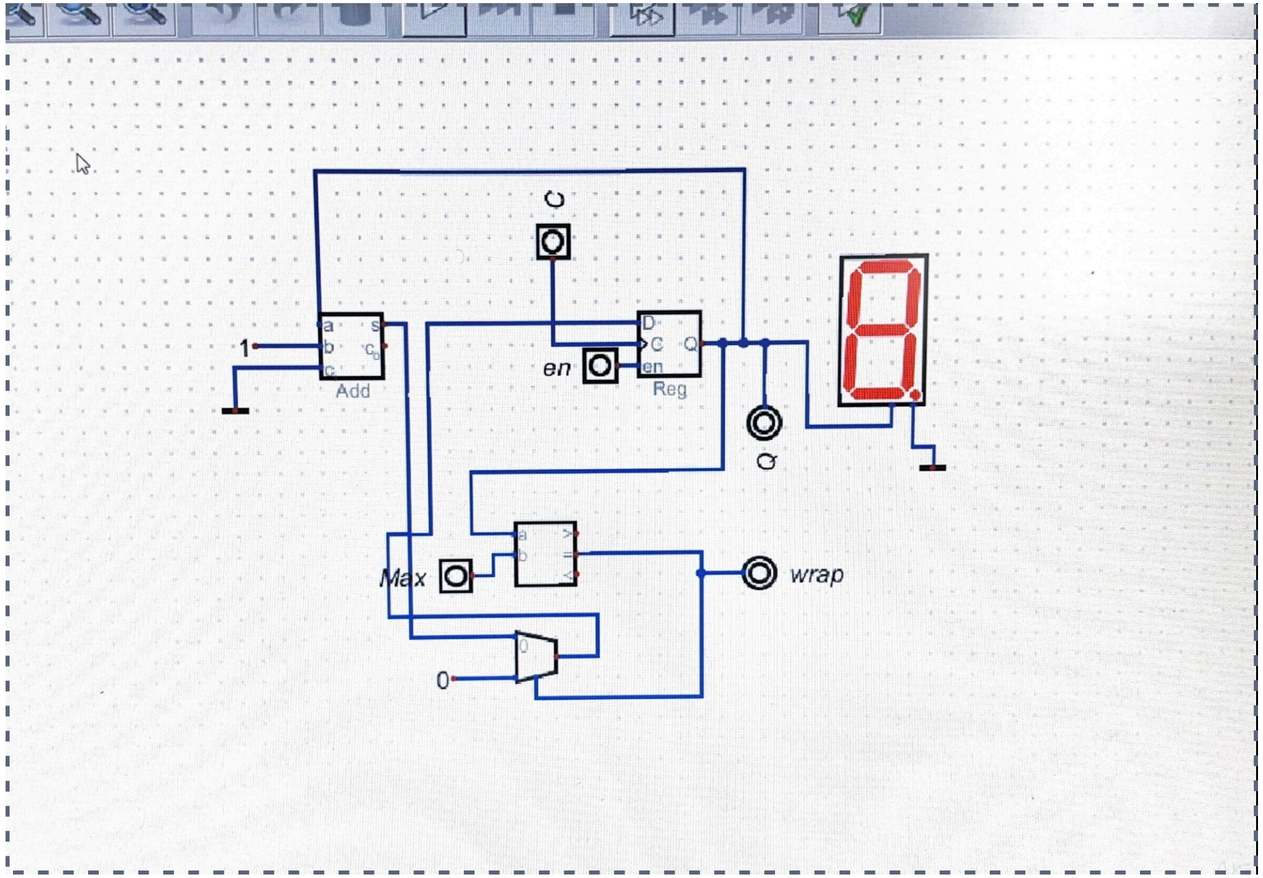
Q	Next Q	wrap
0	1	0
1	2	0
2	3	0
3	4	0
4	5	0
5	0	1

Multiplexer

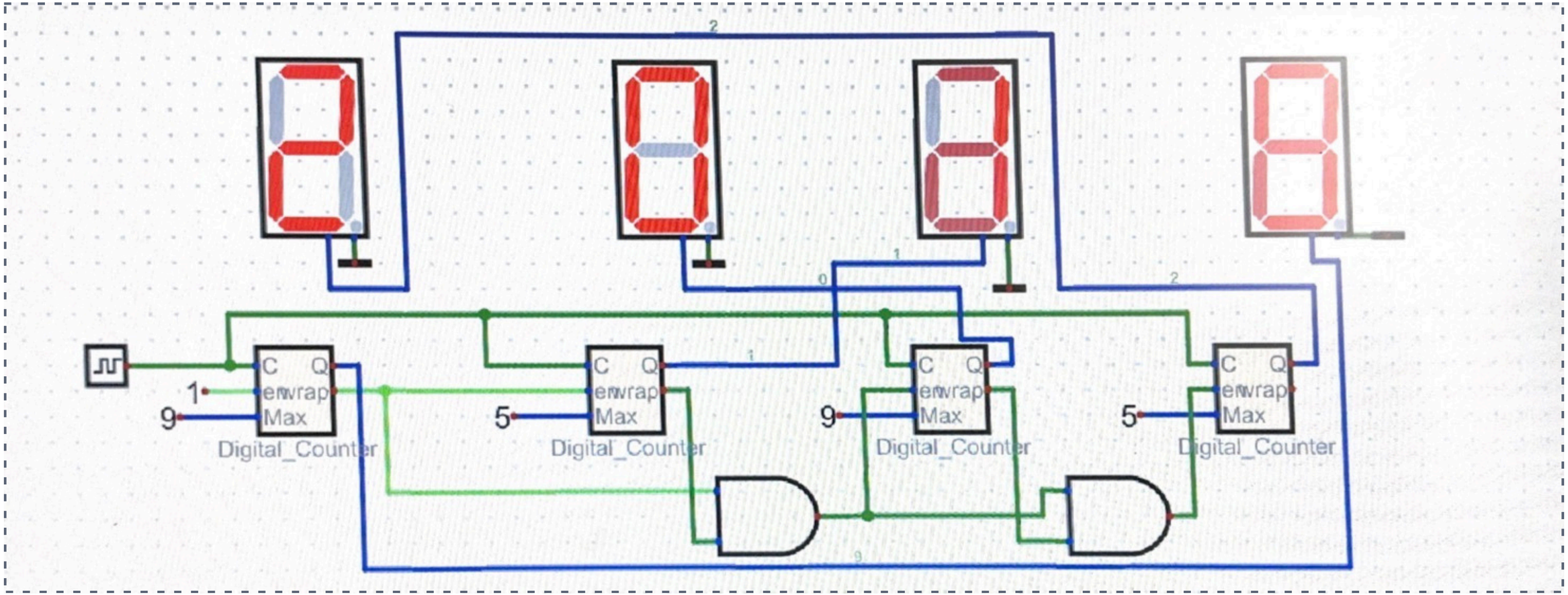
wrap	D (input of register)
0	Q + 1
1	0

Boolean Expressions

Counter	D = Q + 1 if wrap = 0, D = 0 if wrap = 1
wrap	wrap = (Q = Max)
Seconds units	en = 1
Seconds tens	en = wrap1
Minutes units	en = wrap1 · wrap2
Minutes tens	en = wrap1 · wrap2 · wrap3



Logic Circuit Design



Four counter blocks (Max 9, 5, 9, 5), one 1 Hz Clock, two AND gates for the carry chain, four hex displays.